MAGIVOX AI

Data Structures & Algorithms - A to Z

Complexity | Arrays | Linked Lists | Stacks | Queues | Trees | Graphs | Interview

How to use this guide
Study one topic, type the example yourself, practise without copying, then take the MAGIVOX topic test. After all modules, take the final course exam.

Learning outcomes

- Problem Solving Foundation
- Big-O Complexity
- Arrays
- Strings
- Linked Lists
- Stacks

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1. Problem Solving Foundation

What to learn

- Algorithm, input and output
- Dry run
- Pseudocode
- Edge cases
- Time and space complexity

Practice checkpoint

Explain algorithm, input and output in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

2. Big-O Complexity

What to learn

- O(1), O(log n), O(n), O(n log n), O(n^2)
- Best/average/worst case
- Amortized idea

Practice checkpoint

Explain o(1), o(log n), o(n), o(n log n), o(n^2) in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

3. Arrays

What to learn

- Traversal
- Insert/delete concept
- Prefix sums
- Two pointers
- Sliding window

Practice checkpoint

Explain traversal in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

4. Strings

What to learn

- Character frequency
- Palindrome
- Anagram
- Substring patterns

Practice checkpoint

Explain character frequency in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

5. Linked Lists

What to learn

- Singly/doubly linked list
- Insert/delete
- Reverse list
- Fast and slow pointers

Practice checkpoint

Explain singly/doubly linked list in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

6. Stacks

What to learn

- LIFO operations
- Balanced brackets
- Next greater element
- Expression evaluation

Practice checkpoint

Explain lifo operations in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

7. Queues

What to learn

- FIFO operations
- Circular queue
- Deque
- Priority queue idea

Practice checkpoint

Explain fifo operations in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

8. Hashing

What to learn

- Hash table idea
- Set vs Map

- Frequency map
- Collision concept

Practice checkpoint

Explain hash table idea in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

9. Recursion and Backtracking

What to learn

- Base case
- Call stack
- Subsets/permutations
- Backtracking template

Practice checkpoint

Explain base case in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

10. Searching

What to learn

- Linear search
- Binary search
- Binary search on answer

Practice checkpoint

Explain linear search in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

11. Sorting

What to learn

- Bubble/selection/insertion
- Merge sort
- Quick sort
- Stable vs unstable

Practice checkpoint

Explain bubble/selection/insertion in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

12. Trees

What to learn

- Binary tree terminology

- DFS traversals
- BFS level order
- BST operations
- Height and balance

Practice checkpoint

Explain binary tree terminology in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

13. Heaps

What to learn

- Min heap and max heap
- Top K problems
- PriorityQueue usage

Practice checkpoint

Explain min heap and max heap in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

14. Graphs

What to learn

- Vertices and edges
- Adjacency list
- BFS and DFS
- Connected components
- Shortest path overview

Practice checkpoint

Explain vertices and edges in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

15. Dynamic Programming

What to learn

- Overlapping subproblems
- Memoization
- Tabulation
- 1D DP starter problems

Practice checkpoint

Explain overlapping subproblems in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

16. Interview Strategy

What to learn

- Clarify constraints
- Start brute force
- Optimize
- Explain complexity
- Test with sample and edge cases

Practice checkpoint

Explain clarify constraints in your own words. Give one example and one common mistake. Then open the MAGIVOX topic test for this section.

Worked Examples

Binary search

```
while (low <= high) { int mid = low + (high-low)/2; ... }
```

Stack use

```
Push opening brackets; on closing bracket, compare with stack top.
```

BFS idea

```
Use a queue, mark visited, process neighbors level by level.
```

Interview Answer Pattern

Use this structure for technical questions:

- 1. Definition - say what it is in one or two lines.
- 2. Why - explain why it is useful.
- 3. Example - give a small real or code example.
- 4. Difference/common mistake - compare with a related concept when relevant.
- 5. Be truthful - do not claim tools or experience you have not used.

Final Course Checklist

- I can explain the foundation without notes.
- I completed examples by typing/doing them myself.
- I completed topic tests and reviewed wrong answers.
- I completed at least one mini project or practical exercise.
- I can answer common interview questions in simple language.
- I am ready for the MAGIVOX 50-question final exam.